



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

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**SECI1143 – KEBARANGKALIAN STATISTIK & ANALISIS DATA (PROBABILITY
& STATISTICAL DATA ANALYSIS)**

SECTION 03

ASSIGNMENT 4 – CHAPTER 7

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Assignment 4

- (1) (a) The two most popular correlation coefficients are Pearson's product-moment correlation coefficient and Spearman's rho rank correlation coefficient.

(b)	Number of items	Production cost			
	x	y	xy	x^2	y^2
	26	42	1092	676	1764
	44	60	2640	1936	3600
	53	69	3657	2809	4761
	29	47	1363	841	2209
	77	91	7007	5929	8281
	80	98	7840	6400	9604
	20	39	780	400	1521
	40	55	2200	1600	3025
	67	85	5695	4489	7225
	86	104	8944	7396	10816
	17	37	629	289	1369
	61	77	4697	3721	5929
	$\Sigma = 600$	$\Sigma = 804$	$\Sigma = 46544$	$\Sigma = 36486$	$\Sigma = 60104$

$$r = \frac{\Sigma xy - (\Sigma x \Sigma y) / n}{\sqrt{[(\Sigma x^2) - (\Sigma x)^2 / n][(\Sigma y^2) - (\Sigma y)^2 / n]}}$$

$$= \frac{46544 - (600)(804) / 12}{\sqrt{[(36486) - (600)^2 / 12][60104 - (804)^2 / 12]}}$$

$$= 0.998$$

- (c) Based on the correlation coefficient r obtained, we can conclude that there is a very strong positive linear relationship between the number of items produced and the production cost. The value of r being close to 1 indicates that as the number of items produced increases, the production cost also increases in a nearly linear manner.

2. (a)	X	Y	XY	X ²	Y ²
	0.27	2	0.54	0.0729	4
	1.41	3	4.23	1.9881	9
	2.19	3	6.57	4.7961	9
	2.83	6	16.98	8.0089	36
	2.19	4	8.76	4.7961	16
	1.81	2	3.62	3.2761	4
	0.85	1	0.85	0.7225	1
	3.05	5	15.25	9.3025	25
	$\Sigma = 14.6$	$\Sigma = 26$	$\Sigma = 56.8$	$\Sigma = 32.9632$	$\Sigma = 104$

$$r = \frac{56.8 - (14.6)(26)/8}{\sqrt{[32.9632 - (14.6)^2/8][104 - (26)^2/8]}}$$

$$= 0.8424$$

(b) $H_0: \rho = 0$ (No linear correlation) $t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$
 $H_1: \rho \neq 0$ (Linear correlation exists)

$$\alpha = 1 - 0.095 \quad df = 8 - 2$$

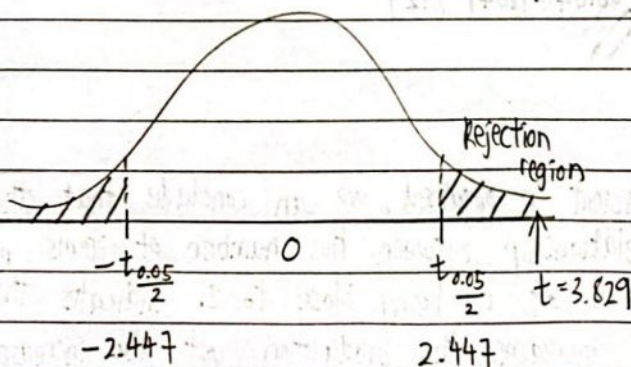
$$\alpha = 0.05 \quad = 6$$

$$= \frac{0.8424}{\sqrt{\frac{1 - (0.8424)^2}{8 - 2}}}$$

$$= \frac{0.8424}{\sqrt{\frac{1 - (0.8424)^2}{8 - 2}}}$$

$$t_{\frac{0.05}{2}, 6} = 2.447$$

$$t = 3.829$$



Since t -statistic (3.829) is greater than the critical t -value (2.447), we reject the null hypothesis, H_0 . There is sufficient evidence of a linear relationship between X and Y (weight of plastic usage) and Y (the size of household) at the 95% confidence level.

2.(c) For a 99% confidence level (two-tailed test), the significance level α is 0.01.

$$\frac{0.01}{2}$$

$$t_{\frac{0.01}{2}, 6} = 3.707 \text{ (refer } t \text{ table)}$$

Since the calculated t -statistic (3.829) is still greater than the critical t -value (3.707), we would still reject the null hypothesis H_0 even at the 99% confidence level.

Thus, increasing the confidence level to 99% does not change the decision. There is still significant evidence to suggest that the X (weight of plastic usage) and Y (the size of household) are ~~are~~ correlated.

3) (a) Engagement score and sentiment score because both are ordinal data.

(b) Number of likes and number of comments because both are ratio variable

Post ID	1	2	3	4	5	6
Engagement score	85	70	90	60	88	75
Sentiment score	4	3	5	2	5	3

Post ID	Rank engagement score	Rank Sentiment score	d_i	d_i^2
1	3	3	0	0
2	5	4.5	0.5	0.25
3	1	1.5	-1.5	2.25
4	6	6	0	0
5	2	1.5	0.5	0.25
6	4	4.5	-0.5	0.25
				$\sum d_i^2 = 1$

$$r_s = 1 - \frac{6(\sum d_i^2)}{n(n^2 - 1)} = 1 - \frac{6(1)}{6(36 - 1)}$$

$$= 0.971$$

(d) (i) Like and comments

Like(x)	Comments(y)	xy	x^2	y^2
150	20	3000	22500	400
100	10	1000	10000	100
200	25	5000	40000	625
80	8	640	6400	64
170	22	3740	28900	484
120	15	1800	14400	225

$$\sum x = 720 \quad \sum y = 100 \quad \sum xy = 15180 \quad \sum x^2 = 122200 \quad \sum y^2 = 1898$$

$$r = \frac{\sum xy - (\sum x \sum y) / n}{\sqrt{[\sum x^2 - (\sum x)^2 / n][\sum y^2 - (\sum y)^2 / n]}}$$

$$= \frac{15180 - (820)(100)/6}{\sqrt{[(122200) - (820)^2/6][1898 - (100)^2/6]}}$$

$$= 0.988$$

(ii) Like and share

Like (x)	share (y)	xy	x ²	y ²
150	30	4500	22500	900
100	20	2000	10000	400
200	40	8000	40000	1600
80	15	1200	6400	225
170	35	5950	28900	1225
120	25	3000	14400	625
$\Sigma x = 820$	$\Sigma y = 165$	$\Sigma xy = 24650$	$\Sigma x^2 = 122200$	$\Sigma y^2 = 4975$

$$r = \frac{\Sigma xy - (\Sigma x \Sigma y)/n}{\sqrt{[(\Sigma x^2) - (\Sigma x)^2/n][(\Sigma y^2) - (\Sigma y)^2/n]}}$$

$$= \frac{24650 - (820)(165)/6}{\sqrt{[(122200) - (820)^2/6][4975 - (165)^2/6]}}$$

$$= 0.997$$

(iii) Engagement score and sentiment score

$$r_s = 0.971$$

- The strongest positive correlation is Like and share, $r = 0.997$
- There is no strongest negative correlation because all r is positive

(e) $H_0: \rho = 0$ $H_1: \rho \neq 0$ $\alpha = 0.05$ $n = 6$ $df = 4$	Engagement score (x)	Post Length (y)	xy	x ²	y ²
	85	200	17000	7225	40000
	70	100	7000	4900	10000
	90	250	22500	8100	62500
	60	150	9000	3600	22500
	98	220	19360	7744	48400
	75	170	13500	5625	32400

$$\sum x = 468$$

$$\sum y = 1100$$

$$\sum xy = 88360 \quad \sum x^2 = 37194 \quad \sum y^2 = 215800$$

$$r = \frac{\sum xy - (\sum x \sum y) / n}{\sqrt{[(\sum x^2) - (\sum x)^2 / n][(\sum y^2) - (\sum y)^2 / n]}}$$

$$r = \frac{88360 - (468)(1100) / 6}{\sqrt{[37194 - (468)^2 / 6][215800 - (1100)^2 / 6]}}$$

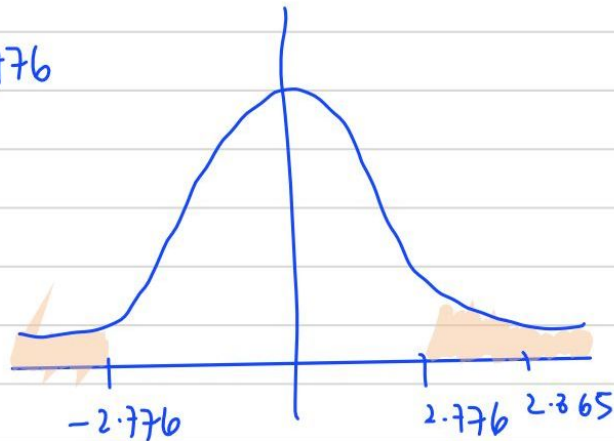
$$= 0.82$$

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

$$= \frac{0.82}{\sqrt{\frac{1-(0.82)^2}{6-2}}}$$

$$= 2.865$$

$$t_{0.025, 4} = 2.776$$



- Since $t = 2.865 > t_{0.025, 4} = 2.776$. We reject the H_0 . There is enough evidence that there is a significant correlation at 0.05 significant level

Question 4

a)

x	y	xy	x ²
95	85	8075	9025
85	95	8075	7225
80	70	5600	6400
70	75	5250	4900
65	70	4550	4225
$\Sigma x = 395$	$\Sigma y = 395$	$\Sigma xy = 31550$	$\Sigma x^2 = 31775$

$$b_1 = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{n}}{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}$$

$$= \frac{31550 - \frac{(395)(395)}{5}}{31775 - \frac{(395)^2}{5}}$$

$$= 0.61$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$= 79 - 0.61(79)$$

$$= 30.81$$

$$\bar{y} = \frac{395}{5} = 79$$

$$\bar{x} = \frac{395}{5} = 79$$

c) $\hat{y} = b_0 + b_1 x$

$$= 30.81 + 0.61x$$

e) $\hat{y} = 30.81 + 0.61(60)$

$$= 67.41$$

f) $SSR = \Sigma (\hat{y} - \bar{y})^2$

$x = 95$ $\hat{y} = 30.81 + 0.61(95)$

$$= 88.76$$

$x = 85$ $\hat{y} = 30.81 + 0.61(85)$

$$= 82.66$$

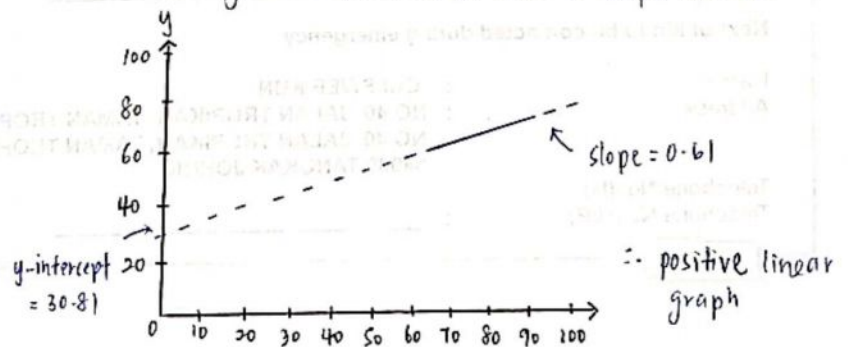
$x = 80$ $\hat{y} = 30.81 + 0.61(80)$

$$= 79.61$$

$x = 70$ $\hat{y} = 30.81 + 0.61(70)$

$$= 73.51$$

d) The graph should be a straight line with a y-intercept of 30.81 and a slope of 0.61.



$x = 65$ $\hat{y} = 30.81 + 0.61(65)$

$$= 70.46$$

$$SSR = (88.76 - 79)^2 + (82.66 - 79)^2 + (79.61 - 79)^2 + (73.51 - 79)^2 + (70.46 - 79)^2$$

$$= 212.10$$

$$SST = \Sigma (y - \bar{y})^2$$

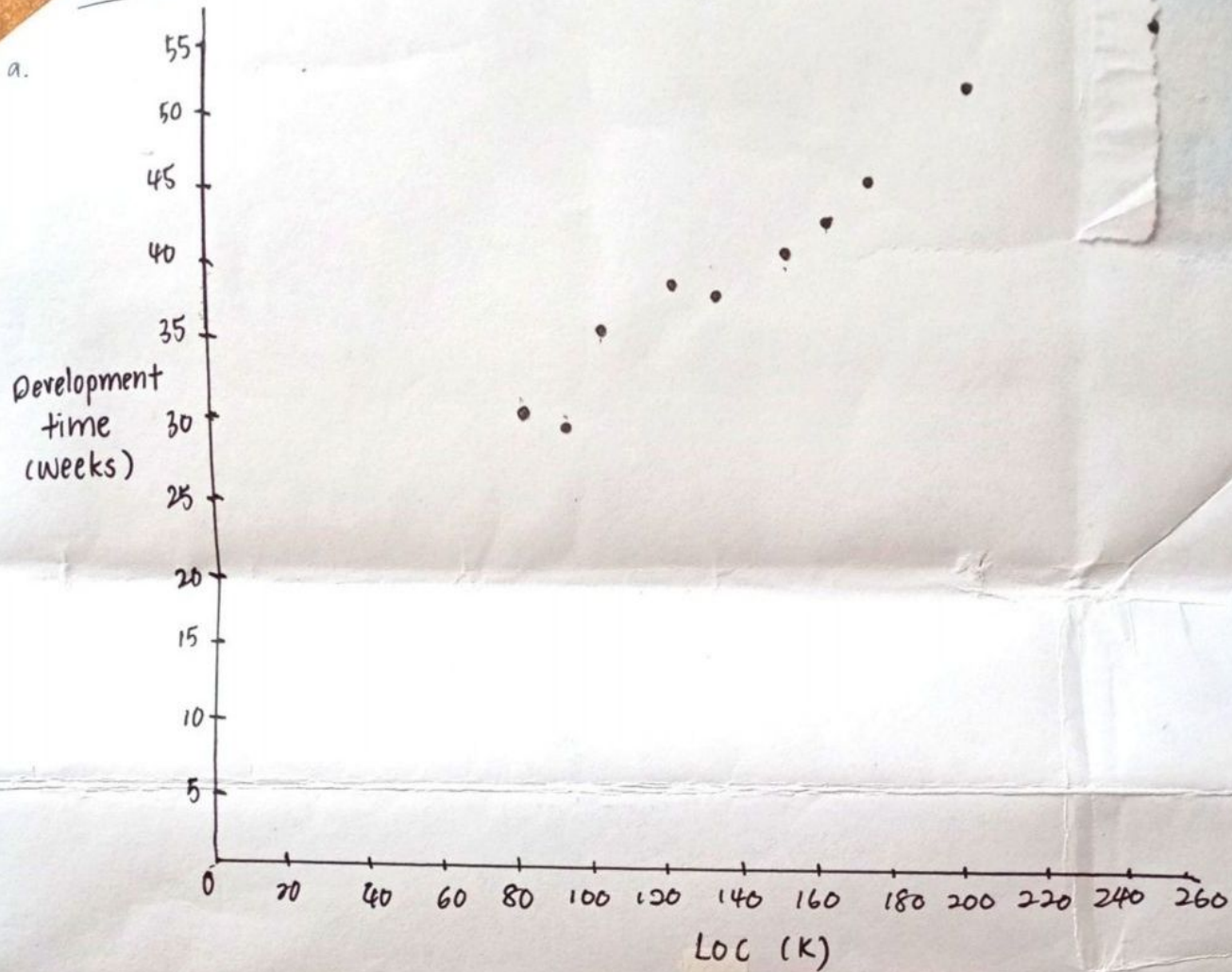
$$= (85 - 79)^2 + (95 - 79)^2 + (70 - 79)^2 + (75 - 79)^2 + (70 - 79)^2$$

$$= 470$$

$$\begin{aligned}
 R^2 &= \frac{SSR}{SST} \\
 &= \frac{212.70}{470} \\
 &= 0.45
 \end{aligned}$$

- 9) Since $0 < R^2 = 0.45 < 1$, regression model explains 45% of the variability in the statistics grades based on the math aptitude test grades are moderate fit. It is a weaker linear relationship as only some statistics grades is explained by variation in math aptitude test.

Question 5



b.

Development time, y	LOC, x	xy	y^2	x^2
40	150	6000	1600	22500
35	100	3500	1225	10000
50	200	10000	2500	40000
30	80	2400	900	6400
45	170	7650	2025	28900
38	120	4560	1444	14400
42	160	6720	1764	25600
28	90	2520	784	8100
55	250	13750	3025	62500
37	130	4810	1369	16900
$\sum y = 400$	$\sum x = 1450$	$\sum xy = 61910$	$\sum y^2 = 16636$	$\sum x^2 = 235300$

$$r = \frac{61910 - \frac{(400)(1450)}{10}}{\sqrt{\left[235300 - \frac{1450^2}{10}\right] \left[16636 - \frac{400^2}{10}\right]}}$$

$$= \frac{3910}{\sqrt{(25050)(636)}}$$

$$= 0.980$$

c. $\hat{y} = b_0 + b_1 x$
 $= 17.380 + 0.156 x$

$$b_1 = \frac{61910 - \frac{400(1450)}{10}}{235300 - \frac{1450^2}{10}}$$

$$= \frac{3910}{25050}$$

$$= 0.156$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$= 40 - 0.156(145)$$

$$= 17.380$$

The intercept (17.380) is the estimated development time when LOC is 0. The slope (0.156) indicates that for each additional 1K lines of code, the development time increases by approximately 0.156 weeks.

d. For a simple linear regression, $R^2 = r^2$.

From (c), $r = 0.980$

$$R^2 = r^2$$

$$= (0.980)^2$$

$$= 0.960$$

$R^2 = 0.960$ means that 96.0% of the variability in the development time can be explained by the linear relationship with LOC. This indicates a very good fit of the regression model to the data.

e. $\bar{y} = b_0 + b_1 \bar{x}$
 $= 17.38 + 0.156(180)$
 $= 45.46$

New development time = 45.46 weeks